

Harnessing Rydberg Atomic Receivers (v5, accepted) — Complete Equation-by-Equation Reference

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1 The three receiver types — overview

The paper compares **three** receiver architectures throughout.

1. **Conventional RF receiver:** classical dipole antenna + LNA superheterodyne chain. Pure baseline, no Rydberg physics.
2. **LO-free Rydberg atomic receiver:** reads the RF field from the Autler–Townes (AT) splitting of the EIT peak directly. Measures amplitude only.

3. **LO-dressed Rydberg atomic receiver:** a strong extra LO RF tone is injected on-resonance with the same Rydberg transition, turning the atoms into a coherent mixer. Measures amplitude *and* phase, and is the most sensitive of the three.

Figure	What it shows / which receiver(s)
Fig. 1	Schematic only (energy levels, LO-free vs LO-dressed setups)
Fig. 2	Noise-chain schematic, all three receivers (block diagram only)
Fig. 3	P_{out} vs $\Delta_c, \Omega_{\text{RF}}$ — LO-free, no link budget yet
Fig. 4	P_{out} heatmap vs $\Delta_c, \mathcal{R}$ — LO-free resolvability
Fig. 5	P_{out} vs $\Delta_c, d_{\text{Tx-Rx}} + \text{SNR}$ — LO-free, full link budget
Fig. 6	Linear dynamic range (LDR) of the LO-dressed receiver
Fig. 7	LDR translated into P_{Rx} bounds vs Ω_{LO} — LO-dressed
Fig. 8	SNR vs distance — all three receivers compared
Fig. 9	Mutual information vs P_{Rx} — all three receivers
Fig. 10	SER vs P_{Rx} — all three receivers

2 Section II: Preliminaries

Eq. (1) — the AT-splitting readout relation

$$\Delta\nu = \begin{cases} \frac{\Omega_{\text{RF}}}{2\pi}, & \text{scanning the coupling laser,} \\ \frac{\Omega_{\text{RF}}}{2\pi} \frac{f_p}{f_c}, & \text{scanning the probe laser.} \end{cases} \quad (1)$$

Terms: $\Delta\nu$ is the frequency separation between the two split EIT peaks that you actually observe on the photodetector. Ω_{RF} is the RF Rabi frequency (what you are trying to measure). f_p, f_c are the probe and coupling laser optical frequencies. **Significance:** this is the foundational readout equation of the whole LO-free concept: it says the splitting you literally see encodes Ω_{RF} directly. Scanning the coupling laser gives the clean, calibration-free form $\Delta\nu = \Omega_{\text{RF}}/2\pi$ (no probe absorption background), which is why it's preferred experimentally. **Not used for Fig. 3 itself** (Fig. 3 plots P_{out} directly against Δ_c , not $\Delta\nu$) — but it is the conceptual justification for why Fig. 3's x -axis is Δ_c at all.

Eq. (2) — Beer–Lambert probe transmission

$$P_{\text{out}}(t) = P_{\text{in}} \exp\{-k_p L \Im(\chi(t))\}. \quad (2)$$

Terms: $P_{\text{out}}(t)$ = transmitted probe laser intensity measured on the photodetector — *this is the quantity plotted on the z -axis of Fig. 3, and of every other P_{out} figure in the paper.* P_{in} = incident probe laser intensity (before the vapor cell). $k_p = 2\pi/\lambda_p$ = probe wavenumber. L = vapor-cell length. $\chi(t)$ = complex electric susceptibility of the atomic medium at time t ; only its imaginary part matters here because $\Im(\chi)$ is what controls absorption (the real part would control dispersion/phase shift, not amplitude loss). **Significance:** this is a literal, standard Beer–Lambert absorption law, with the medium's absorption coefficient written as $k_p \Im(\chi)$. **This is the single formula that turns the quantum-mechanical coherence into the number plotted in Fig. 3** — every value in our 3D surface is computed by plugging a value of $\Im(\chi)$ into this exact exponential. If Fig. 3 doesn't match the paper, the error must live either (a) upstream, in what $\chi(t)$ or $\Im(\chi)$ actually evaluates to (wrong Hamiltonian, wrong steady state, wrong parameters), or

(b) in the constants k_p , L , P_{in} fed into this exact exponential. There is no other place for a bug to hide in this equation itself — it’s a one-line formula.

Eq. (3) — P_{in} in terms of the probe Rabi frequency

$$P_{\text{in}} = \frac{\pi}{2Z_0} \left(\frac{d\Omega_p\hbar}{2\sqrt{\wp_{12}}} \right)^2. \quad (3)$$

Terms: d = probe beam $1/e^2$ diameter. Ω_p = probe Rabi frequency. $\hbar$ = reduced Planck’s constant. $\wp_{12}$ = dipole moment of the $|1\rangle$ – $|2\rangle$ transition (**note:** as written, $\wp_{12}$ is used *under a square root* here — confirms $\wp_{12}$ is defined as an already-squared quantity, i.e. $\wp_{12} = (2.5ea_0)^2$ as stated later in the paper, not $\wp_{12} = 2.5ea_0$). Z_0 = free-space impedance. **Significance:** this equation is the physical link between the probe beam’s intensity/power and its Rabi frequency (via the beam diameter d , which sets the field’s spatial concentration). **This is the exact equation at the center of the known P_{in} /diameter inconsistency:** plugging in the paper’s own stated $d = 0.76$ mm and $\Omega_p/2\pi = 8$ MHz gives $P_{\text{in}} = 39.08 \mu\text{W}$, not the $20.7 \mu\text{W}$ the paper also states directly in the same parameter list. **This equation is not directly used to compute Fig. 3’s P_{out} surface** (we use the stated $P_{\text{in}} = 20.7 \mu\text{W}$ directly in Eq. 2, and the stated $\Omega_p/2\pi = 8$ MHz directly in the Hamiltonian, Eq. 6) — but it is the reason those two paper-stated numbers can never both be exactly self-consistent, which is one confirmed, real source of an overall multiplicative scale mismatch (though NOT of any shape/sharpness mismatch, since Eq. 3 only sets an overall constant, not how P_{out} depends on Δ_c or Ω_{RF}).

Eq. (4) — susceptibility in terms of the coherence

$$\chi(t) = C_0 \rho_{21}(t), \quad C_0 = \frac{-2N_0\wp_{12}}{\epsilon_0\hbar\Omega_p}. \quad (4)$$

Terms: $\rho_{21}(t)$ = the $|1\rangle$ – $|2\rangle$ coherence, an off-diagonal element of the atomic density matrix — *the single quantity that the entire quantum-mechanical part of the model (Hamiltonian + Lindblad + steady state) exists to compute*. C_0 = a real, negative constant bundling N_0 (atomic density), $\wp_{12}$ (dipole moment), ϵ_0 (vacuum permittivity), $\hbar$, and Ω_p . N_0 = total atomic number density in the vapor cell. **Significance:** this equation is the direct bridge between “quantum physics” (ρ_{21} , computed via QuTiP) and “classical optics” (χ , fed into Beer–Lambert, Eq. 2). **C_0 is exactly where $N_0 = 4.89 \times 10^{16} \text{ m}^{-3}$ enters the whole calculation** — since C_0 is a simple linear prefactor, doubling N_0 exactly doubles $\Im(\chi)$ for the same ρ_{21} , which (through Eq. 2’s exponential) can massively change P_{out} ’s absolute scale and apparent sharpness. This is the equation responsible for the already-confirmed “high- N_0 compression” effect: because N_0 is large, small physically-correct changes in ρ_{21} (from changing Δ_c or Ω_{RF} by a little) get amplified into large changes in $\Im(\chi)$, and then further amplified exponentially by Eq. 2 — producing a sharper, narrower-looking P_{out} curve than a naive linear intuition would suggest.

Eq. (5) — the master equation

$$\dot{\rho} = \frac{\partial \rho}{\partial t} = -\frac{j}{\hbar}[\mathbf{H}, \rho] + \mathcal{L}. \quad (5)$$

Terms: ρ = the full 4×4 density matrix of the 4-level atomic system. $\mathbf{H}$ = the system Hamiltonian (Eq. 6). $[\mathbf{H}, \rho] = \mathbf{H}\rho - \rho\mathbf{H}$ = the commutator, generating coherent (unitary) evolution. $\mathcal{L}$ = the Lindblad dissipator (Eq. 7), generating incoherent (spontaneous-emission) evolution. $j = \sqrt{-1}$ (engineering notation for the imaginary unit, used throughout the paper instead of i). **Significance:**

this is the standard Lindblad master equation for an open quantum system — the general-purpose equation that `qutip.steadystate()` solves (by construction) when you pass it $\mathbf{H}$ and the collapse operators. **There is no ambiguity or possible bug in this equation itself** — QuTiP’s steady-state solver implements exactly this, given a correct $\mathbf{H}$ and a correct set of collapse operators. Any discrepancy must trace back to $\mathbf{H}$ (Eq. 6) or the collapse operators (Eq. 7), not to this equation.

Eq. (6) — the 4-level Hamiltonian

$$\mathbf{H} = \frac{\hbar}{2} \begin{bmatrix} 0 & \Omega_p & 0 & 0 \\ \Omega_p & -2\Delta_p & \Omega_c & 0 \\ 0 & \Omega_c & -2(\Delta_p + \Delta_c) & \Omega \\ 0 & 0 & \Omega & -2(\Delta_p + \Delta_c + \Delta_{\text{RF}}) \end{bmatrix}. \quad (6)$$

Terms: basis states $|1\rangle, |2\rangle, |3\rangle, |4\rangle$ correspond to Cs levels $6S_{1/2}, 6P_{3/2}, 47D_{5/2}, 48P_{3/2}$. Ω_p = probe Rabi frequency (couples $|1\rangle \leftrightarrow |2\rangle$). Ω_c = coupling Rabi frequency (couples $|2\rangle \leftrightarrow |3\rangle$). Ω (bare, unsubscripted) = the $|3\rangle \leftrightarrow |4\rangle$ coupling — **the paper explicitly states in the text immediately after this equation that “the Rabi frequency Ω in (6) differs for LO-free and LO-dressed Rydberg atomic receivers.”** For the LO-free receiver (Fig. 3, 4, 5), $\Omega = \Omega_{\text{RF}}$ directly (confirmed by Eq. 8’s own use of Ω_{RF} in exactly this matrix slot). For the LO-dressed receiver (Fig. 6, 7), $\Omega = \Omega_{\text{total}} = \Omega_{\text{LO}} + e^{j(2\pi\Delta f t + \Delta\phi)}\Omega_{\text{RF}}$ (a complex phasor, defined later at Eq. 26/28 context). $\Delta_p, \Delta_c, \Delta_{\text{RF}}$ = detunings of the probe, coupling, and RF fields from their respective transitions. **Significance:** this is *the* Hamiltonian used for every single QuTiP steady-state solve in this whole reproduction (Fig. 3 through Fig. 7 all reuse this exact 4-level structure, changing only which detunings are swept and what value goes in the Ω slot). **For debugging Fig. 3 specifically:** confirm your code’s matrix uses $-\Delta_p$ (not $-2\Delta_p$) on the diagonal if your Hamiltonian is defined directly in angular-frequency units without an explicit $\hbar/2$ prefactor (i.e., if you write $\mathbf{H} = \dots$ without dividing by $\hbar$ afterward) — dividing the paper’s $\frac{\hbar}{2} \times (-2\Delta_p)$ by $\hbar$ gives exactly $-\Delta_p$, which is what a Hamiltonian-in-frequency-units convention (the usual QuTiP convention for a Hermitian generator with $\hbar = 1$) should use. This has already been verified correct in this project’s code.

Eq. (7) — the Lindblad dissipator

$$\mathcal{L} = \begin{bmatrix} \gamma_2\rho_{22} & -\gamma_{12}\rho_{12} & -\gamma_{13}\rho_{13} & -\gamma_{14}\rho_{14} \\ -\gamma_{21}\rho_{21} & \gamma_3\rho_{33} - \gamma_2\rho_{22} & -\gamma_{23}\rho_{23} & -\gamma_{24}\rho_{24} \\ -\gamma_{31}\rho_{31} & -\gamma_{32}\rho_{32} & \gamma_4\rho_{44} - \gamma_3\rho_{33} & -\gamma_{34}\rho_{34} \\ -\gamma_{41}\rho_{41} & -\gamma_{42}\rho_{42} & -\gamma_{43}\rho_{43} & -\gamma_4\rho_{44} \end{bmatrix}, \quad \gamma_{ij} = \frac{\gamma_i + \gamma_j}{2}. \quad (7)$$

Terms: $\gamma_2, \gamma_3, \gamma_4$ = spontaneous-emission (population decay) rates out of levels 2, 3, 4 respectively ($\gamma_1 = 0$ since $|1\rangle$ is the ground state and cannot decay further). $\gamma_{ij} = (\gamma_i + \gamma_j)/2$ = the dephasing rate of coherence ρ_{ij} , a standard relation for decay-induced dephasing. **Significance:** this matrix describes a simple *cascade decay* chain: $|2\rangle \rightarrow |1\rangle$ at rate γ_2 , $|3\rangle \rightarrow |2\rangle$ at rate γ_3 , $|4\rangle \rightarrow |3\rangle$ at rate γ_4 — population only ever flows *downward*. This exact structure is generated automatically by three QuTiP collapse operators $\sqrt{\gamma_2}|1\rangle\langle 2|$, $\sqrt{\gamma_3}|2\rangle\langle 3|$, $\sqrt{\gamma_4}|3\rangle\langle 4|$ — if your code’s collapse operators are anything other than exactly these three (e.g., missing one, wrong ket/bra order, or an extra spurious dephasing-only operator), the resulting steady state will differ from the paper’s model, silently. **This has already been verified correct** in this project’s Fig. 3 code by direct comparison of the collapse-operator list against this matrix structure.

Eq. (8) — closed-form steady-state coherence (weak-probe limit)

$$\rho_{21} = \frac{-j(\Omega_p/2)}{\gamma_{21} - j\Delta_p - \frac{(\Omega_c/2)^2}{\gamma_{31} - j(\Delta_p + \Delta_c) - \frac{(\Omega_{\text{RF}}/2)^2}{\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{\text{RF}})}}}. \quad (8)$$

Terms: a nested continued fraction in $\Omega_p, \Omega_c, \Omega_{\text{RF}}$ and the three detunings, using the same γ_{ij} dephasing rates as Eq. 7. **Significance:** this is the paper’s own *analytic* alternative to a full QuTiP steady-state solve, derived (Appendix A) under the weak-probe approximation $\rho_{11} \approx 1$, $\rho_{22}, \rho_{33}, \rho_{44} \approx 0$. **Verified this session to be numerically pathological in Fig. 3’s actual operating regime:** at $\Delta_c = 0$ and $\Omega_{\text{RF}} = 0$ it gives unphysical gain ($P_{\text{out}} > P_{\text{in}}$), and for any $\Omega_{\text{RF}} \gtrsim 0.5$ MHz it collapses to numerically zero and stays flat — because $\gamma_4/2\pi = 1.7$ kHz is so tiny that the innermost term $(\Omega_{\text{RF}}/2)^2/\gamma_{41}$ explodes almost immediately. Tried three different sign conventions (flip the outer j , flip all j ’s, flip only the detuning signs) — none produce a sane curve. **Do not use Eq. 8 to try to reproduce or debug Fig. 3** — the paper’s own Fig. 3 was generated with full QuTiP (stated directly in Sec. V-A: “We employ the QuTiP toolkit for simulating the quantum system”), not this closed form. This equation is only useful for physical intuition and for deriving other analytic results elsewhere in the paper (e.g., the LO-dressed Appendix C derivations).

Eq. (9) — Maxwell–Boltzmann velocity distribution

$$g(v_z) = \frac{1}{\sqrt{2\pi}\sigma_v} e^{-v_z^2/(2\sigma_v^2)}. \quad (9)$$

Terms: v_z = atomic velocity component along the laser propagation axis. $\sigma_v = \sqrt{k_B T/m_{\text{Rb}}}$ = the thermal velocity spread (standard deviation), with k_B = Boltzmann constant, T = vapor temperature, m_{Rb} = atomic mass. **Significance:** standard 1-D thermal velocity distribution for a room-temperature atomic vapor — the starting point for modeling Doppler broadening.

Eq. (10) — Doppler-shifted detunings

$$\Delta_p(v_z) = \Delta_p - \frac{2\pi}{\lambda_p} v_z, \quad \Delta_c(v_z) = \Delta_c + \frac{2\pi}{\lambda_c} v_z. \quad (10)$$

Terms: λ_p, λ_c = probe/coupling laser wavelengths. The opposite signs on the two Doppler-shift terms come directly from the probe and coupling lasers being **counter-propagating** (stated explicitly in the text) — an atom moving toward one beam is moving away from the other. **Significance:** this is how a single atomic velocity v_z maps onto an *effective*, atom-specific pair of detunings that then get substituted into Eq. 8 (or the full Hamiltonian) before averaging. **Directly relevant to the Fig. 3 debugging question:** this project deliberately does *not* apply Eq. 10/11 to Fig. 3, because a real convergence test showed that properly-converged Doppler averaging completely erases the AT splitting (a $\sim 0\%$ dip at $\Delta_c = 0$), which contradicts the real published Fig. 3 (which clearly shows resolved splitting). The paper’s own cited source for this exact kind of EIT-AT spectrum plot explicitly states it uses a **cold-atom model** (no Doppler averaging) for its own analogous figure. If you want to re-litigate this decision as part of your deep dive, this is the equation pair to revisit — but the convergence-test evidence against including it is strong.

Eq. (11) — Doppler-averaged coherence

$$\bar{\rho}_{21}(\Delta_p) = \int_{-\infty}^{+\infty} \rho_{21}(v_z) g(v_z) dv_z. \quad (11)$$

Terms: $\rho_{21}(v_z)$ = the coherence computed (via Eq. 8, or a full solve) using the velocity-shifted detunings from Eq. 10, for one specific atomic velocity class v_z . The integral averages over the full thermal velocity distribution $g(v_z)$ from Eq. 9. **Significance:** this is the final Doppler-averaging step — in principle, the “correct” ρ_{21} to feed into Eq. 4/2 for a warm vapor. As noted under Eq. 10, this project’s convergence testing showed this integral needs $\gtrsim 400$ velocity-class samples to converge (given the single-photon Doppler width is ~ 158 MHz versus the $\sim$ MHz-scale EIT features), and a properly-converged version erases the AT splitting entirely — which is why Fig. 3 in this project uses the *unaveraged*, single-velocity-class ($v_z = 0$, i.e. cold-atom) result instead.

3 Section III-A: LO-free Rydberg atomic receiver model

Eq. (12) — RF Rabi frequency from the field

$$\Omega_{\text{RF}} = \frac{|E_{\text{RF}}| \wp_{\text{RF}}}{\hbar}. \quad (12)$$

Terms: E_{RF} = incident RF electric field amplitude. $\wp_{\text{RF}}$ = dipole moment of the $|3\rangle$ – $|4\rangle$ (Rydberg–Rydberg) transition, stated as $\wp_{\text{RF}} = -1443.459 ea_0$. **Significance:** the fundamental field-to-Rabi-frequency conversion, used any time a distance or transmit power needs to become an Ω_{RF} value to feed into the quantum model (Fig. 5, Fig. 7). Not used by Fig. 3 itself (Fig. 3 sweeps Ω_{RF} directly as an independent axis, with no link budget involved).

Eq. (13) — LO-free signal model and received power definition

$$y = \sqrt{P_{\text{Rx}}} hx + n_{\text{ex}}, \quad P_{\text{Rx}} = \frac{|E_{\text{RF}}|^2}{2Z_0}. \quad (13)$$

Terms: $x \sim \mathcal{CN}(0,1)$ = unit-power complex Gaussian transmitted symbol. $h \sim \mathcal{CN}(0,1)$ = complex Gaussian channel gain. n_{ex} = extrinsic (BBR) noise. P_{Rx} = received RF power, defined here directly from the field amplitude. **Significance:** sets up the standard wireless-communications signal model wrapping around the Rydberg physics — not used for Fig. 3 (a pure physics figure with no wireless-link elements).

Eq. (14) — Friis transmission equation

$$P_{\text{Rx}} = P_{\text{Tx}} G_{\text{Tx}} G_{\text{Rx}} \left(\frac{\lambda}{4\pi d_{\text{Tx-Rx}}} \right)^2. \quad (14)$$

Terms: $P_{\text{Tx}}, G_{\text{Tx}}$ = transmit power and Tx antenna gain. G_{Rx} = classical receive antenna gain. $d_{\text{Tx-Rx}}$ = link distance. λ = RF wavelength. **Significance:** the standard classical link-budget formula, quoted here explicitly as *not directly applicable* to the Rydberg receiver, since “there is no metallic antenna and hence no meaningful classical gain G_{Rx} ” — this sets up why Eq. 15–17 are needed instead.

Eq. (15) — received power flux density

$$S_{\text{Rx}} = \frac{P_{\text{Tx}} G_{\text{Tx}}}{4\pi d_{\text{Tx-Rx}}^2}. \quad (15)$$

Terms: S_{Rx} = power flux density (units W/m²) at the receiver location. **Significance:** this is Eq. 14 with the G_{Rx} and wavelength-squared factor stripped out — i.e. just the raw free-space power density arriving at the receiver’s location, independent of what kind of receiver sits there.

Eq. (16) — power via effective aperture

$$P_{\text{Rx}} = S_{\text{Rx}} A_{\text{eff}}. \quad (16)$$

Terms: A_{eff} = the Rydberg receiver’s effective aperture (Eq. 17). **Significance:** replaces the classical antenna-gain-based power-capture picture with an aperture-based one — appropriate since the Rydberg atoms “sense” the field via dipole coupling rather than guiding currents through a metallic structure.

Eq. (17) — effective aperture

$$A_{\text{eff}} = \frac{2Z_0 N_{\text{atoms}} \wp_{\text{RF}}^2 \omega_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}}, \quad \omega_{\text{RF}} = 2\pi f_{\text{RF}}. \quad (17)$$

Terms: N_{atoms} = number of atoms effectively contributing to the reception (**the paper never gives this a numeric value or an explicit formula anywhere** — confirmed by exhaustive search of the paper and its three cited references; this is a genuine, real gap, not something you’re missing in your reading). Γ_{FWHM} = the “intrinsic EIT linewidth” (also never given one single, unambiguous closed-form value that’s consistent across every figure that uses it — see the per-figure notes near the end of this document). **Significance:** derived in Appendix B from first principles (scattering-rate $\times$ energy-per-photon arguments) — confirms A_{eff} scales with N_{atoms} and the transition’s own properties, explicitly *not* the physical cross-sectional area of the vapor cell. Not used by Fig. 3 (no link budget there).

Eq. (18) — LO-free observation model, recast

$$z = \frac{\wp_{\text{RF}}}{\hbar} |E_{\text{RF}} \hbar x + n_{\text{Ry}}|. \quad (18)$$

Terms: z = the scalar LO-free observation (physically, an estimated Rabi frequency). n_{Ry} = the lumped noise term covering all relevant noise sources (detailed later in Eq. 46). **Significance:** recasts Eq. 13’s signal model into the actual physical quantity the LO-free receiver outputs — a magnitude (Rabi-frequency) estimate, not a complex baseband voltage like a conventional receiver would produce.

Eq. (19) — piecewise-constant RF amplitude assumption

$$x(t) \approx x[k], \quad t \in [kT_m, (k+1)T_m]. \quad (19)$$

Terms: T_m = one measurement/scan window duration. **Significance:** formalizes the assumption that the incident RF amplitude doesn’t change appreciably during one full coupling-laser scan — needed to justify treating one scan as producing one valid observation.

Eq. (20) — coupling-laser scan trajectory

$$\Delta_c(t) = \Delta_{c,\min} + \frac{\Delta_{c,\max} - \Delta_{c,\min}}{T_m}(t - kT_m), \quad t \in [kT_m, (k+1)T_m]. \quad (20)$$

Terms: $\Delta_{c,\min}, \Delta_{c,\max}$ = the scanned detuning range's endpoints. **Significance:** describes the coupling laser being linearly swept (a ramp) across a fixed detuning window once per measurement window — this is exactly the kind of sweep Fig. 3's Δ_c -axis represents conceptually, though Fig. 3 itself just plots the steady-state result at each Δ_c , not a literal time-domain scan.

Eq. (21) — continuous-time photodetector output

$$v_k(t) = R_{\text{PD}}P_{\text{out}}(\Delta_c(t)) + n_{\text{RY},k}(t), \quad t \in [kT_m, (k+1)T_m]. \quad (21)$$

Terms: R_{PD} = photodetector responsivity. $n_{\text{RY},k}(t)$ = the relevant noise contributions during scan k . **Significance:** what the PD literally outputs during one scan — the real $P_{\text{out}}(\Delta_c)$ trace (exactly what Fig. 3 plots, one Ω_{RF} slice at a time) plus noise.

Eq. (22) — sampled photodetector output

$$v_k[m] = R_{\text{PD}}P_{\text{out}}(\Delta_c[m]) + n_{\text{RY},k}[m]. \quad (22)$$

Terms: $m = 0, \dots, M-1$ indexes M discrete detuning sample points within one scan. **Significance:** the discrete-time version of Eq. 21 — this is literally what one column of a digitized Fig. 3-style sweep looks like in practice.

Eq. (23) — the LO-free discrete observation

$$z[k] \triangleq \Omega_{\text{RF}}[k] = \mathcal{E} \left(v_k[m]_{m=0}^{M-1} \right). \quad (23)$$

Terms: $\mathcal{E}(\cdot)$ = an AT-splitting extraction procedure (e.g. peak-separation detection, or model-based fitting). **Significance:** formalizes the LO-free receiver's whole job as a function $\mathcal{E}$ that turns one entire scan trace (i.e., one Fig. 3-style curve) into a single scalar Ω_{RF} estimate. This is also why the LO-free symbol rate is capped at $R_s \leq 1/T_m$ — fundamentally, it needs one whole scan per observation.

4 Section III-B: LO-dressed Rydberg atomic receiver model

Eq. (24) — LO and RF field definitions

$$E_{\text{LO}} = A_{\text{LO}} \cos(2\pi f_{\text{LO}}t + \phi_{\text{LO}}), \quad E_{\text{RF}} = A_{\text{RF}} \cos(2\pi f_{\text{RF}}t + \phi_{\text{RF}}). \quad (24\text{a}, 24\text{b})$$

Terms: $A_{\text{LO}}, A_{\text{RF}}$ = amplitudes ($A_{(\cdot)} = |E_{(\cdot)}|$). $f_{\text{LO}}, f_{\text{RF}}$ = frequencies. $\phi_{\text{LO}}, \phi_{\text{RF}}$ = phases. Define $\Delta f = f_{\text{LO}} - f_{\text{RF}}$, $\Delta\phi = \phi_{\text{LO}} - \phi_{\text{RF}}$. **Significance:** simply defines the two real physical RF tones (the deliberately-injected LO, and the actual signal to be measured) that get superposed in the LO-dressed architecture.

Eq. (25) — exact field superposition

$$\begin{aligned} E_{\text{total}} &= E_{\text{LO}} + E_{\text{RF}} \\ &= \sqrt{A_{\text{LO}}^2 + A_{\text{RF}}^2 + 2A_{\text{LO}}A_{\text{RF}}\cos(2\pi\Delta f t + \Delta\phi)} \cos(2\pi f_{\text{LO}}t + \phi_{\text{LO}}). \end{aligned} \quad (25)$$

Significance: the exact sum of the two cosines, rewritten as a single carrier at f_{LO} with a slowly-time-varying *envelope* amplitude that depends on Δf and $\Delta\phi$. This is a pure trigonometric identity, not an approximation, and shows the physical origin of envelope/beat detection in the LO-dressed scheme.

Eq. (26) — small-signal (Taylor) approximation

$$E_{\text{total}} \approx [A_{\text{LO}} + A_{\text{RF}}\cos(2\pi\Delta f t + \Delta\phi)] \cos(2\pi f_{\text{LO}}t + \phi_{\text{LO}}), \quad (A_{\text{RF}}/A_{\text{LO}} \ll 1). \quad (26)$$

Significance: a second-order Taylor expansion of Eq. 25 around $A_{\text{RF}}/A_{\text{LO}} = 0$, valid whenever the RF signal is much weaker than the injected LO (the intended operating regime). Physically: the carrier at f_{LO} has its amplitude modulated by a slow cosine at Δf — this is literally amplitude-modulation-style envelope detection, and is the direct justification for approximating Ω_{total} 's magnitude the way Fig. 6/7's build does.

Eq. (27) — closed-form ρ_{21} at resonance, metastable approximation

$$\rho_{21}|_{(\Delta_p, \Delta_c, \Delta_{\text{LO}}, \gamma_3, \gamma_4)=0} = \frac{j\gamma_2\Omega_p|\Omega_{\text{total}}|^2}{\gamma_2^2|\Omega_{\text{total}}|^2 + 2\Omega_p^2(\Omega_c^2 + \Omega_p^2 + |\Omega_{\text{total}}|^2)}. \quad (27)$$

Terms: evaluated at full resonance ($\Delta_p = \Delta_c = \Delta_{\text{LO}} = 0$) and under the approximation $\gamma_3 = \gamma_4 = 0$ (justified since states 3, 4 are metastable Rydberg states with much longer lifetimes than state 2, i.e. $\gamma_3, \gamma_4 \ll \gamma_2$). $\Omega_{\text{total}} = \Omega_{\text{LO}} + e^{j(2\pi\Delta f t + \Delta\phi)}\Omega_{\text{RF}}$ (complex phasor, defined just before this equation in the text). **Significance:** the LO-dressed analogue of Eq. 8 — another closed-form steady state, but valid specifically at exact resonance with the two shortest-lived-state decay rates dropped. This is the starting point for the entire LDR/distortion derivation in Appendix C (Eq. 75–83) and Sec. IV-B (Eq. 53–58).

Eq. (28) — practical LO-dressed output

$$P_{\text{out}}(t) = \bar{P}_0 + \kappa\Omega_{\text{RF}}\cos(2\pi\Delta f t + \Delta\phi). \quad (28)$$

Terms: $\bar{P}_0 = P_{\text{in}}e^{-\alpha}e^{\alpha\Lambda(\Omega_{\text{LO}}, \Gamma)}$ is the average (DC) probe transmission, set purely by the LO bias Ω_{LO} . $\alpha = k_p L C_0 \bar{A}$ is the absorption coefficient of the probe laser, with $\bar{A} = \gamma_2\Omega_p/(\gamma_2^2 + 2\Omega_p^2) =$ the amplitude of the (simplified, three-level) EIT spectrum. $\Lambda(a, b) = b^2/(b^2 + a^2)$ is a normalized Lorentzian function of variable a with HWHM b . $\Gamma = \Omega_p\sqrt{2(\Omega_c^2 + \Omega_p^2)/(2\Omega_p^2 + \gamma_2^2)}$ is the HWHM of a four-level Lorentzian (defined again, identically, at Eq. 78). $\kappa = \alpha\bar{P}_0\kappa_\rho$ is the **intrinsic gain coefficient**, with $\kappa_\rho = \partial\Lambda(\Omega_{\text{LO}}, \Gamma)/\partial\Omega_{\text{LO}}$. **Significance:** this is the central result of the LO-dressed model — it says $P_{\text{out}}(t)$ oscillates *linearly* in Ω_{RF} around a DC bias set entirely by Ω_{LO} . κ is the single most important LO-dressed design figure of merit: the whole point of choosing a good Ω_{LO} is to maximize $|\kappa|$ (done at Eq. 79). **Not used by Fig. 3** (Fig. 3 is LO-free, no Ω_{LO} involved at all).

Eq. (29) — AC-coupled readout

$$P_{\text{readout}}(t) = P_{\text{out}}(t) - \bar{P}_0 = |P(\Delta f)| \cos(2\pi\Delta f t + \Delta\phi). \quad (29)$$

Terms: $|P(\Delta f)|$ = the amplitude of the single-sided Fourier spectrum of $P_{\text{out}}(t)$ evaluated at Δf — i.e. literally what a spectrum analyzer or lock-in amplifier reads out. **Significance:** subtracting off the DC bias $\bar{P}_0$ isolates the useful oscillating signal, whose amplitude directly encodes Ω_{RF} via Eq. 30.

Eq. (30) — RF Rabi frequency from the Fourier amplitude

$$\Omega_{\text{RF}} = \frac{|P(\Delta f)|}{|\kappa|}. \quad (30)$$

Significance: the actual measurement recipe for the LO-dressed receiver — measure the Fourier-spectrum peak at Δf , divide by the known/calibrated κ , and you have Ω_{RF} (and hence, via Eq. 12, the RF field amplitude $A_{\text{RF}} = \hbar\Omega_{\text{RF}}/\wp_{\text{RF}}$). Both $|P(\Delta f)|$ and $|\kappa|$ can be obtained directly from the optical spectrum, which is why this technique reduces electric-field measurement to a purely optical-frequency readout.

5 Section III-C: Noise modeling

Fig. 2 provides an explicit block-diagram (Tx → RF signal → vapor cell → PD → LNA → LIA → observation) for all three receivers side by side, labeling exactly where each noise term enters and with what multiplicative gain. The equations below are the formulas behind each labeled block.

Eq. (31) — AC photocurrent

$$I_{\text{AC}} = D \left| \kappa \right| \frac{\wp_{\text{RF}}}{\hbar} \sqrt{P_{\text{Rx}}} + D \left| \kappa \right| \frac{\wp_{\text{RF}}}{\hbar} n_{\text{ex}} + D n_{\text{PSN}} \quad (31)$$

Terms: D = photodiode responsivity (A/W). n_{PSN} = photon shot noise contribution. **Significance:** converts the optical AC signal (amplitude $|\kappa|\Omega_{\text{RF}}$, per Eq. 28) into an electrical photocurrent, carrying both the useful signal and the extrinsic + shot-noise contributions through the same responsivity D .

Eq. (32) — electrical AC power

$$P_{\text{eAC}} = I_{\text{AC}}^2 R_L. \quad (32)$$

Terms: R_L = PD output load impedance. **Significance:** simple $I^2 R$ conversion from photocurrent to electrical power, at the point right after the photodetector.

Eq. (33) — LNA output power

$$P_{\text{LNA}} = G_{\text{LNA}} P_{\text{eAC}} + \sigma_{\text{TN}}^2. \quad (33)$$

Terms: G_{LNA} = low-noise-amplifier gain. σ_{TN}^2 = thermal (Johnson–Nyquist) noise power, added at this stage. **Significance:** amplifies the PD’s electrical output and adds the amplifier’s own thermal noise floor.

Eq. (34) — full LO-dressed observation, with channel

$$z^{\text{LO}} = \sqrt{P_{\text{Rx}} G_{\text{LNA}} R_L} D |\kappa| \frac{\wp_{\text{RF}}}{\hbar} h x + n_{\text{Ry,LO}}. \quad (34)$$

Significance: the complete end-to-end LO-dressed observation model, folding the wireless channel h , the transmitted symbol x , and every gain stage (PD responsivity, LNA gain, load resistance) into one expression, with $n_{\text{Ry,LO}}$ collecting all noise (detailed at Eq. 47). This is the LO-dressed counterpart of Eq. 18.

Eq. (35) — noise-equivalent field (NEF)

$$\text{NEF} = \frac{|E_{\text{RF}}|_{\min}}{\sqrt{B}}. \quad (35)$$

Terms: $|E_{\text{RF}}|_{\min}$ = minimum detectable field amplitude. B = measurement bandwidth. **Significance:** a bandwidth-normalized noise floor expressed in the receiver’s natural “field” units ($\text{V/m}/\sqrt{\text{Hz}}$) — a smaller NEF means a more sensitive receiver. This is the standard figure of merit quoted throughout the Rydberg-sensor literature.

Eq. (36) — extrinsic NEF

$$\text{NEF}_{\text{ex}} = \sqrt{\frac{16\pi f_{\text{RF}}^2}{3\epsilon_0 c^3} \Theta(f_{\text{RF}}, T)}. \quad (36)$$

Terms: c = speed of light. **Significance:** the external (BBR + vacuum-fluctuation) noise floor, expressed as an NEF — this is a fundamental limit that no free-space receiver, Rydberg or classical, can beat.

Eq. (37) — modified Callen–Welton spectral factor

$$\Theta(f_{\text{RF}}, T) = \hbar f_{\text{RF}} \begin{cases} \frac{1}{2} n_{\text{th}}(f_{\text{RF}}, T) + \frac{1}{2}, & \text{homodyne,} \\ 2n_{\text{th}}(f_{\text{RF}}, T) + 1, & \text{heterodyne.} \end{cases} \quad (37)$$

Significance: distinguishes the noise contribution depending on whether the detection is homodyne or heterodyne — this factor folds both the thermal (BBR) and quantum vacuum contributions into one expression via n_{th} (Eq. 38).

Eq. (38) — thermal photon occupation number

$$n_{\text{th}}(f_{\text{RF}}, T) = \frac{1}{\exp(\hbar f_{\text{RF}}/k_B T) - 1}. \quad (38)$$

Significance: the standard Bose–Einstein thermal occupation number at frequency f_{RF} and temperature T — the number of thermal photons per mode.

Eq. (39) — extrinsic noise power

$$\sigma_{\text{ex}}^2 = \frac{\text{NEF}_{\text{ex}}^2}{2Z_0} A_{\text{eff}} B. \quad (39)$$

Significance: converts the extrinsic NEF (a field-domain quantity) into an actual noise *power* at the receiver, using the effective aperture A_{eff} (Eq. 17) to do the field→power conversion and B to set the bandwidth.

Eq. (40) — standard quantum limit (SQL) field

$$\frac{E_{\text{SQL}}}{\sqrt{\text{Hz}}} = \frac{\hbar}{\wp_{\text{RF}} \sqrt{N_{\text{atoms}} T_2}}. \quad (40)$$

Terms: T_2 = the EIT coherence time. **Significance:** the fundamental atomic-shot-noise-limited sensitivity floor — scales down as $1/\sqrt{N_{\text{atoms}}}$, the standard quantum-limit scaling for N_{atoms} independent quantum probes.

Eq. (41) — quantum-projection-noise (QPN) power

$$\sigma_{\text{QPN}}^2 = \left(\frac{E_{\text{SQL}}}{\sqrt{\text{Hz}}} \right)^2 \frac{A_{\text{eff}} B}{2Z_0}. \quad (41)$$

Significance: same field-to-power conversion pattern as Eq. 39, but starting from the SQL field instead of the extrinsic NEF — this is the fundamental atomic-physics noise contribution (as opposed to the BBR-driven extrinsic noise).

Eq. (42) — average photocurrent from probe power

$$\bar{I} = \frac{\eta_{\text{eff}} e}{2\pi \hbar f_p} |P_{\text{out}}|. \quad (42)$$

Terms: $\eta_{\text{eff}} \in [0, 1]$ = PD quantum efficiency. e = elementary charge. f_p = probe laser frequency. **Significance:** standard photodiode relation — converts incident optical power $|P_{\text{out}}|$ into average photocurrent via the quantum efficiency and photon energy $\hbar f_p$.

Eq. (43) — photon shot noise (PSN) power

$$\sigma_{\text{PSN}}^2 = 2eB\bar{I}. \quad (43)$$

Significance: the standard shot-noise formula for a photocurrent $\bar{I}$ over bandwidth B — an intrinsic, unavoidable consequence of the discreteness of photon detection.

Eq. (44) — thermal (Johnson–Nyquist) noise power

$$\sigma_{\text{TN}}^2 = \begin{cases} 4k_B T B, & \text{TIA front-end,} \\ F k_B T B, & \text{LNA front-end.} \end{cases} \quad (44)$$

Terms: F = LNA noise factor. **Significance:** the electronic thermal noise floor, with two different forms depending on whether the front end is a transimpedance amplifier (dominated by feedback-resistor Johnson noise) or a matched LNA (dominated by the noise factor F).

Eq. (45) — LO-free small-signal transduction gain

$$G_{\text{Ry}} = \frac{\partial P_{\text{out}}}{\partial E_{\text{RF}}}. \quad (45)$$

Significance: defines the LO-free receiver's own intrinsic gain — the local slope of the probe transmission with respect to the incident RF field. This is deliberately analogous to κ (Eq. 28) for the LO-dressed case, letting the two architectures be compared apples-to-apples in the noise formulas below.

Eq. (46) — total LO-free noise power

$$\sigma_{\text{Ry}}^2 = G_{\text{Ry}}^2 D^2 G_{\text{LNA}}^2 \sigma_{\text{ex}}^2 + D^2 G_{\text{LNA}}^2 \sigma_{\text{QPN}}^2 + G_{\text{LNA}}^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \quad (46)$$

Significance: sums all four noise contributions (extrinsic, QPN, PSN, TN), each referred forward through the correct chain of gains up to the final observation point. Note the extrinsic term is scaled by G_{Ry}^2 , which the text calls “modest,” since the LO-free architecture “operates on the natural slope of an EIT doublet.”

Eq. (47) — total LO-dressed noise power

$$\sigma_{\text{Ry,LO}}^2 = \kappa^2 D^2 G_{\text{LNA}}^2 \sigma_{\text{ex}}^2 + D^2 G_{\text{LNA}}^2 \sigma_{\text{QPN}}^2 + G_{\text{LNA}}^2 \sigma_{\text{PSN}}^2 + \sigma_{\text{TN}}^2. \quad (47)$$

Significance: the single most important noise-comparison equation in the paper — structurally identical to Eq. 46, but the extrinsic term is scaled by κ^2 instead of G_{Ry}^2 . Since κ can be one to two orders of magnitude larger than G_{Ry} (stated directly in the text right after Eq. 47), this single substitution is the quantitative origin of the LO-dressed receiver’s SNR advantage seen later in Fig. 8/9/10.

6 Section IV-A: SNR performance

Eq. (48) — the dimensionless resolvability ratio

$$\mathcal{R} \equiv \frac{\Omega_{\text{RF}}}{\Gamma_{\text{FWHM}}}. \quad (48)$$

Significance: the single dimensionless figure of merit for LO-free AT-splitting resolvability — how large the RF coupling is relative to the intrinsic linewidth. **This is exactly Fig. 4’s y -axis.** Achieving $\mathcal{R} \gg 1$ is stated as a necessary condition for clear splitting and high-fidelity LO-free sensing.

Eq. (49) — Fisher-information penalty

$$G(\mathcal{R}) = \frac{\mathcal{R}^2}{1 + \frac{1}{2\mathcal{R}^2}}. \quad (49)$$

Significance: derived rigorously in Appendix D from Poisson statistics and the Cramér–Rao bound (not an ad-hoc fudge factor) — quantifies the information penalty/resolvability loss of trying to estimate Ω_{RF} from an AT-split spectrum. Collapses toward 0 as $\mathcal{R} \rightarrow 0$ (matching the qualitative statement that below $\mathcal{R} = 1$ the splitting is unobservable), approaches $\mathcal{R}^2$ for $\mathcal{R} \gg 1$.

Eq. (50) — LO-free SNR

$$\text{SNR}_{\text{Ry}} = \frac{P_{\text{Rx}} |h|^2}{\sigma_{\text{Ry}}^2} \left(\frac{\wp_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}} \right)^2 G(\mathcal{R}). \quad (50)$$

Significance: integrates the classical wireless link budget ($P_{\text{Rx}} |h|^2 / \sigma_{\text{Ry}}^2$) with the RF-to-atomic conversion gain $(\wp_{\text{RF}} / \hbar \Gamma_{\text{FWHM}})^2$ and the resolvability penalty $G(\mathcal{R})$ — these three multiplicative factors are individually dimensionless-consistent, so $G(\mathcal{R})$ only reshapes the numerical value, not the physical units.

Eq. (51) — LO-free sensitivity floor

$$E_{\min}^{\text{strong}} = \frac{\hbar \Gamma_{\text{EIT}}}{\wp_{\text{RF}}} \left(\frac{2Z_0 \sigma_{\text{Ry}}^2}{A_{\text{eff}} |\hbar|^2} \right)^{1/4}. \quad (51)$$

Significance: obtained by setting $\text{SNR}_{\text{Ry}} = 1$ in Eq. 50 and solving for the minimum detectable field — the paper reports a computed value of $38.15 \mu\text{V}/\text{cm}/\sqrt{\text{Hz}}$, cross-checked as consistent with existing literature.

Eq. (52) — LO-dressed SNR

$$\text{SNR}_{\text{Ry,LO}} = \frac{G_{\text{LNA}} R_L D^2 \kappa^2 \frac{\wp_{\text{RF}}^2}{\hbar^2} P_{\text{Rx}} |\hbar|^2}{\sigma_{\text{Ry,LO}}^2}. \quad (52)$$

Significance: the LO-dressed SNR expression, built from the signal model (Eq. 34) and total noise (Eq. 47) — note κ^2 appears explicitly in the numerator (signal power) as well as inside $\sigma_{\text{Ry,LO}}^2$'s extrinsic term (Eq. 47), which is why Eq. 61 below is a more physically revealing rearrangement.

7 Section IV-B: Distortion effect / linear dynamic range

Eq. (53) — restructured probe transmission

$$P_{\text{out}} = P_{\text{in}} e^{-\alpha} e^{\alpha \Lambda(\Omega_{\text{total}}, \Lambda)}. \quad (53)$$

Significance: rewrites P_{out} (same physical quantity as Eq. 2/28) explicitly as a function of the Lorentzian Λ evaluated at the *total* Rabi frequency Ω_{total} — the starting point for the Taylor-expansion-based distortion analysis that follows.

Eq. (54) — Taylor expansion of the Lorentzian

$$\Lambda(\Omega_{\text{total}}, \Lambda) = \Lambda_0 + \Lambda'_0 \delta\Omega + \frac{1}{2} \Lambda''_0 \delta\Omega^2 + \frac{1}{6} \Lambda'''_0 \delta\Omega^3 + \dots \quad (54)$$

Terms: $\delta\Omega = \Omega_{\text{total}} - \Omega_{\text{LO}}$. $\Lambda'_0 = \frac{-2\Gamma^2 \Omega_{\text{LO}}}{(\Gamma^2 + \Omega_{\text{LO}}^2)^2}$, $\Lambda''_0 = \frac{-2\Gamma^2(\Gamma^2 - 3\Omega_{\text{LO}}^2)}{(\Gamma^2 + \Omega_{\text{LO}}^2)^3}$ (explicit derivative formulas, evaluated at the bias point Ω_{LO}). **Significance:** a standard Taylor expansion around the LO bias point — this is exactly the mathematical machinery used to separate P_{out} into a desired linear term plus 2nd- and 3rd-order distortion terms in Eq. 55.

Eq. (55) — P_{out} split into desired + distortion terms

$$P_{\text{out}} = \bar{P}_0 + \underbrace{\kappa \Omega_{\text{RF}} \cos \theta}_{\text{desired}} + \underbrace{\frac{1}{4} \alpha \bar{P}_0 \Lambda''_0 \Omega_{\text{RF}}^2 (1 + \cos 2\theta)}_{\text{2nd-order distortion}} + \underbrace{\frac{1}{6} \alpha \bar{P}_0 \Lambda'''_0 \Omega_{\text{RF}}^3 \cos^3 \theta}_{\text{3rd-order distortion}}. \quad (55)$$

Terms: $\theta = 2\pi \Delta f t + \Delta\phi$. **Significance:** substituting Eq. 54 into P_{out} and retaining terms up to third order in Ω_{RF} produces exactly three physically distinct terms: the wanted linear signal (oscillating at Δf , amplitude $\propto \kappa$), a second-harmonic distortion term (oscillating at $2\Delta f$, amplitude $\propto \Lambda''_0$), and a third-harmonic distortion term (oscillating at $3\Delta f$, amplitude $\propto \Lambda'''_0$). This is the direct mathematical origin of the “linear dynamic range” concept: as Ω_{RF} grows, the distortion terms (which grow as $\Omega_{\text{RF}}^2, \Omega_{\text{RF}}^3$) eventually catch up to and overwhelm the desired term (which only grows as Ω_{RF}^1).

Eq. (56) — total-harmonic-distortion (THD) ratio

$$\frac{P(\Delta f)}{P(2\Delta f)} = \frac{\kappa\Omega_{\text{RF}}}{\frac{1}{4}\alpha\bar{P}_0\Lambda_0''\Omega_{\text{RF}}^2} \leq \epsilon. \quad (56)$$

Terms: ϵ = the total harmonic distortion (THD) tolerance. **Note:** the paper’s own equation number/label here reads “ $P(\Delta f)/P(3\Delta f)$ ” in one place, which appears to be a typo, since the actual formula shown uses the second-harmonic term $P(2\Delta f) = \frac{1}{4}\alpha\bar{P}_0\Lambda_0''\Omega_{\text{RF}}^2$ defined in the sentence immediately preceding it — treat this as comparing fundamental to *second*-harmonic power, consistent with Eq. 55’s “2nd-order distortion” label. **Significance:** defines the design criterion — keep the second-harmonic-to-fundamental power ratio below some tolerance ϵ — that Eq. 57 directly solves for Ω_{RF} .

Eq. (57) — second-order LDR upper bound (VERIFY: contains ϵ)

$$\Omega_{\text{RF},\text{max}}'' = \frac{4\epsilon|\Lambda_0'|}{|\Lambda_0''|}. \quad (57)$$

Terms: ϵ = the same THD tolerance from Eq. 56 — **confirmed by directly re-reading the rendered paper page: this equation DOES contain ϵ .** It is not simply $4|\Lambda_0'|/|\Lambda_0''|$ — that ratio is only the coefficient that ϵ multiplies. **Significance:** solving Eq. 56 for Ω_{RF} gives the maximum RF Rabi frequency you can apply before second-order distortion exceeds your chosen tolerance ϵ . **Practical caveat:** $\Lambda_0'' \rightarrow 0$ exactly at the true optimal bias point $\Omega_{\text{LO,opt}}$ (Eq. 79), which makes this bound blow up (divide by zero) there — this is not a flaw in the equation, it’s the direct mathematical statement that “second-order distortion vanishes at the optimum,” which is precisely why Eq. 58 (third-order) is needed at that specific bias point.

Eq. (58) — third-order LDR upper bound (used at the true optimum)

$$\Omega_{\text{RF},\text{max}}''' = \sqrt{\frac{6\epsilon|\Lambda_0'|}{|\Lambda_0'''|}}. \quad (58)$$

Terms: Λ_0''' = the third derivative of Λ with respect to Ω_{LO} at the bias point (not given an explicit closed form in the paper text shown, but obtainable by differentiating Λ_0'' ’s formula once more, or numerically from real data). **Significance:** the governing LDR bound specifically *at* the optimal bias $\Omega_{\text{LO,opt}}$, where second-order distortion has vanished and third-order distortion becomes the limiting factor. **The paper never assigns ϵ a numeric value anywhere** — it calls ϵ “the designer-specified tolerance,” explicitly framing it as a free engineering choice, not a physical constant. Any numeric LDR width computed from Eq. 57 or Eq. 58 therefore requires choosing your own ϵ and labeling it as such.

8 Section IV-C / V-B: mutual information

Eq. (59) — LO-free mutual information

$$\mathcal{I}(z; x) = \frac{1}{2} \ln \left(\frac{\pi(\mathfrak{r} + 1)}{2} \right) - \ln I_0(\mathfrak{r}) + \mathfrak{r} \frac{I_1(\mathfrak{r})}{I_0(\mathfrak{r})}. \quad (59)$$

Terms: I_0, I_1 = modified Bessel functions of the first kind, orders 0 and 1. $\mathfrak{r}$ = the Rician factor, approximated by $\mathfrak{r} \approx \text{SNR}_{\text{Ry}}$. **Significance:** the theoretical mutual information for the LO-free receiver’s signal model (Eq. 18), which behaves like a Rician-distributed amplitude observation — this is exactly what Fig. 9’s green “LO-free” curve plots.

Eq. (60) — LO-dressed mutual information

$$\mathcal{I}(z^{\text{LO}}; x) = e^{1/\text{SNR}_{\text{Ry,LO}}} E_1(1/\text{SNR}_{\text{Ry,LO}}) \ln 2. \quad (60)$$

Terms: $E_1(x) = \int_x^\infty \frac{e^{-t}}{t} dt$ is the exponential integral function. **Significance:** the LO-dressed mutual information, based on the signal model in Eq. 34 (a complex Gaussian observation, unlike the LO-free receiver’s real-amplitude-only observation) — this is what Fig. 9’s orange “LO-dressed” curves plot.

Eq. (61) — restructured LO-dressed SNR

$$\text{SNR}_{\text{Ry,LO}} = \frac{R_L \frac{\sigma_{\text{RF}}^2}{h^2} P_{\text{Rx}} |h|^2}{G_{\text{LNA}} \sigma_{\text{ex}}^2 + \frac{G_{\text{LNA}} \sigma_{\text{QPN}}^2}{\kappa^2} + \frac{G_{\text{LNA}} \sigma_{\text{PSN}}^2}{D^2 \kappa^2} + \frac{\sigma_{\text{TN}}^2}{D^2 \kappa^2 G_{\text{LNA}}}}. \quad (61)$$

Significance: algebraically identical to Eq. 52, but rearranged to make the mechanism explicit: the common extrinsic (BBR) noise σ_{ex}^2 is *not* divided by κ^2 (same κ -dependence as the signal itself, so it doesn’t help), while QPN, PSN, and TN all *are* divided by κ^2 — explaining precisely why the LO-dressed advantage is concentrated in non-BBR-limited operating regimes, not a violation of any shared noise floor.

9 Appendix A: solving the master equation

Eq. (62) — the ρ_{21} commutator element

$$[\mathbf{H}, \boldsymbol{\rho}]_{21} = \frac{\hbar \Omega_p}{2} (\rho_{11} - \rho_{22}) - \hbar \Delta_p \rho_{21} + \frac{\hbar \Omega_c}{2} \rho_{31}. \quad (62)$$

Significance: the explicit $(2, 1)$ matrix element of the commutator $[\mathbf{H}, \boldsymbol{\rho}]$, computed directly from Eq. 6’s Hamiltonian using the general rule $[\mathbf{H}, \boldsymbol{\rho}]_{ij} = \sum_k H_{ik} \rho_{kj} - \rho_{ik} H_{kj}$ — this is the first concrete step toward deriving Eq. 8.

Eq. (63) — ρ_{21} evolution equation

$$\dot{\rho}_{21} = -j \frac{\Omega_p}{2} (\rho_{11} - \rho_{22}) + j \Delta_p \rho_{21} - j \frac{\Omega_c}{2} \rho_{31} - \gamma_{21} \rho_{21}. \quad (63)$$

Significance: substitutes Eq. 62 into the master equation (Eq. 5) and adds the dephasing term $-\gamma_{21} \rho_{21}$ from the Lindblad part (Eq. 7) — the first of three coupled differential equations (with Eq. 64a/b) needed to solve for the full steady state.

Eq. (64) — ρ_{31} and ρ_{41} evolution equations

$$\dot{\rho}_{31} = -j \frac{\Omega_c}{2} (\rho_{21} - \rho_{32}) + j (\Delta_p + \Delta_c) \rho_{31} - j \frac{\Omega_{\text{RF}}}{2} \rho_{41} - \gamma_{31} \rho_{31}, \quad (64a)$$

$$\dot{\rho}_{41} = -j \frac{\Omega_{\text{RF}}}{2} (\rho_{31} - \rho_{42}) + j (\Delta_p + \Delta_c + \Delta_{\text{RF}}) \rho_{41} - \gamma_{41} \rho_{41}. \quad (64b)$$

Significance: the analogous evolution equations for the two other relevant coherences ρ_{31}, ρ_{41} , obtained the same way as Eq. 63 — together, Eq. 63/64a/64b form the full coupled system that needs to be solved (at steady state) to get ρ_{21} .

Eq. (65) — steady-state equations after approximations

$$(\gamma_{21} - j\Delta_p)\rho_{21} = -j\frac{\Omega_p}{2}(\rho_{11} - \rho_{22}) - j\frac{\Omega_c}{2}\rho_{31}, \quad (65a)$$

$$[\gamma_{31} - j(\Delta_p + \Delta_c)]\rho_{31} = -j\frac{\Omega_c}{2}\rho_{21} - j\frac{\Omega_{\text{RF}}}{2}\rho_{41}, \quad (65b)$$

$$[\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{\text{RF}})]\rho_{41} = -j\frac{\Omega_{\text{RF}}}{2}\rho_{31}. \quad (65c)$$

Significance: sets $\dot{\rho}_{21} = \dot{\rho}_{31} = \dot{\rho}_{41} = 0$ (the steady-state condition) in Eq. 63/64a/64b, and applies two stated approximations: (i) $\rho_{11} \approx 1$ (ground-state population probability), so $\rho_{22}, \rho_{33}, \rho_{44}$ are small and neglected; (ii) $\rho_{32}, \rho_{42} \rightarrow 0$ (their influence on ρ_{21} is assumed marginal). This produces a clean, linear, coupled 3-equation system.

Eq. (66) — ρ_{31} eliminated

$$\left[\gamma_{31} - j(\Delta_p + \Delta_c) - \frac{(\Omega_{\text{RF}}/2)^2}{\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{\text{RF}})} \right] \rho_{31} = -j\frac{\Omega_c}{2}\rho_{21}. \quad (66)$$

Significance: substitutes Eq. 65c into Eq. 65b to eliminate ρ_{41} , leaving a single equation relating ρ_{31} to ρ_{21} — the “innermost” layer of the continued fraction that will appear in Eq. 8.

Eq. (67) — ρ_{21} fully resolved (immediately gives Eq. 8)

$$(\gamma_{21} - j\Delta_p)\rho_{21} = -j\frac{\Omega_p}{2}(\rho_{11} - \rho_{22}) - j\frac{\Omega_c}{2} \left(\frac{-j\frac{\Omega_c}{2}}{\gamma_{31} - j(\Delta_p + \Delta_c) - \frac{(\Omega_{\text{RF}}/2)^2}{\gamma_{41} - j(\Delta_p + \Delta_c + \Delta_{\text{RF}})}} \rho_{21} \right). \quad (67)$$

Significance: substitutes Eq. 66 into Eq. 65a. Letting $\rho_{11} \rightarrow 1, \rho_{22} \rightarrow 0$ (weak-probe limit) and solving for ρ_{21} produces exactly Eq. 8 — this equation is the last algebraic step of that derivation, shown explicitly so the full chain from the Hamiltonian to the closed form is traceable.

10 Appendix B: deriving the effective aperture A_{eff}

Eq. (68) — steady-state excited-state population

$$\rho_{ee} = \frac{(\Omega_{\text{RF}}/\Gamma_{\text{FWHM}})^2}{1 + 4(\Delta_c/\Gamma_{\text{FWHM}})^2 + 2(\Omega_{\text{RF}}/\Gamma_{\text{FWHM}})^2}. \quad (68)$$

Terms: ρ_{ee} = the excited-state (Rydberg) population fraction, cited from a standard two-level-atom saturation formula [Steck, Rb87 D-line data]. **Significance:** models how strongly the atoms are driven by the RF field, in a simplified two-level picture, as a function of the ratio $\Omega_{\text{RF}}/\Gamma_{\text{FWHM}}$ and detuning $\Delta_c/\Gamma_{\text{FWHM}}$ — the starting point for computing a scattering rate.

Eq. (69) — total scattering rate

$$\bar{R} = \Gamma_{\text{FWHM}}\rho_{ee}. \quad (69)$$

Significance: the rate at which the atom absorbs and re-emits RF photons — population fraction times the natural linewidth (standard two-level scattering-rate relation).

Eq. (70) — near-resonant, weak-drive scaling

$$\bar{R} = \frac{\Omega_{\text{RF}}^2}{\Gamma_{\text{FWHM}}} = \frac{\wp_{\text{RF}}^2 E_{\text{RF}}^2}{\hbar^2 \Gamma_{\text{FWHM}}}, \quad (\Delta_c \approx 0, \Omega_{\text{RF}} \ll \Gamma_{\text{FWHM}}). \quad (70)$$

Significance: simplifies Eq. 68/69 in the specific limit relevant to weak RF fields at line center, and substitutes $\Omega_{\text{RF}} = \wp_{\text{RF}} E_{\text{RF}}/\hbar$ (Eq. 12) to express the scattering rate directly in terms of the field.

Eq. (71) — power absorbed per atom

$$P_{\text{atom}} = \hbar \omega_{\text{RF}} \bar{R} = \frac{\wp_{\text{RF}}^2 E_{\text{RF}}^2 \omega_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}}. \quad (71)$$

Significance: each absorption–re-emission event removes one photon’s worth of energy $\hbar \omega_{\text{RF}}$ from the field — multiplying by the scattering rate $\bar{R}$ gives the power drawn from the field by a single atom.

Eq. (72) — total power absorbed by all atoms

$$P_{\text{total}} = N_{\text{atoms}} P_{\text{atom}} = \frac{N_{\text{atoms}} \wp_{\text{RF}}^2 E_{\text{RF}}^2 \omega_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}}. \quad (72)$$

Significance: simply scales the single-atom result by N_{atoms} , assuming independent, non-interacting atoms (already justified via the blockade/Rydberg-fraction argument discussed under Fig. 4/Sec. V-A).

Eq. (73) — matching to a flux $\times$ aperture form

$$\frac{E_{\text{RF}}^2}{2Z_0} A_{\text{eff}} = \frac{N_{\text{atoms}} \wp_{\text{RF}}^2 E_{\text{RF}}^2 \omega_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}}. \quad (73)$$

Significance: defines the effective aperture A_{eff} implicitly, by demanding that (power flux density) $\times A_{\text{eff}}$ reproduce the total absorbed power computed in Eq. 72 — the standard way an “aperture” is defined for any receiver.

Eq. (74) — effective aperture, final form

$$A_{\text{eff}} = \frac{2Z_0 N_{\text{atoms}} \wp_{\text{RF}}^2 \omega_{\text{RF}}}{\hbar \Gamma_{\text{FWHM}}}. \quad (74)$$

Significance: solving Eq. 73 for A_{eff} reproduces Eq. 17 exactly — confirms the two are the same formula, one derived from first principles here, one just stated earlier in the main text.

11 Appendix C: deriving the LO-dressed optimum

Eq. (75) — restructured $\Im[\rho_{21}]$

$$\Im[\rho_{21}(\Omega_{\text{RF}})] = \bar{A}[1 - \Lambda(\Omega_{\text{LO}}, \Gamma) - \kappa_{\rho} \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)]. \quad (75)$$

Significance: rewrites Eq. 27’s $\Im[\rho_{21}]$ using the Lorentzian Λ notation, exposing $\kappa_{\rho} = \partial \Lambda(\Omega_{\text{LO}}, \Gamma)/\partial \Omega_{\text{LO}}$ (the intrinsic gain coefficient associated with the coherence itself, before the Beer–Lambert exponential is applied) as a clean, separate factor.

Eq. (76) — $\Im[\rho_{21}]$ at $\Omega_{\text{RF}} = 0$, versus Δ_c

$$\Im[\rho_{21}(\Delta_c)]|_{(\Delta_p, \Delta_{\text{LO}}, \gamma_3, \gamma_4, \Omega_{\text{RF}}, \Omega_{\text{LO}})=0} = \bar{A} - \bar{A} \Lambda(\Delta_c, \Gamma_{\text{HWHM}}^{(3)}). \quad (76)$$

Significance: the special case with no RF/LO field at all — recovers the plain three-level EIT lineshape as a function of Δ_c alone, expressed via the three-level HWHM $\Gamma_{\text{HWHM}}^{(3)}$ (Eq. 77). This is a useful cross-check: it is a completely independent derivation of the basic (no-RF) EIT peak shape, separate from the 4-level Fig. 3 machinery.

Eq. (77) — three-level HWHM

$$\Gamma_{\text{HWHM}}^{(3)} = \frac{\Omega_c^2 + \Omega_p^2}{2\sqrt{\gamma_2^2 + 2\Omega_p^2}}. \quad (77)$$

Significance: the half-width at half-maximum of the *three*-level (no RF/Rydberg-coupling) EIT spectrum — purely a function of the two laser Rabi frequencies and the intermediate-state decay rate γ_2 . **Caution, already established in this project:** this quantity, used alone, is *not* the correct Γ_{FWHM} for Fig. 4/Eq. 48’s LO-free normalization (that candidate was tested and found to predict AT splitting at $\mathcal{R} \approx 0.3\text{--}0.4$, well below the paper’s own stated $\mathcal{R} = 1$ threshold).

Eq. (78) — four-level Γ

$$\Gamma = \frac{2\sqrt{2}\Omega_p}{\sqrt{\Omega_c^2 + \Omega_p^2}} \Gamma_{\text{HWHM}}^{(3)}. \quad (78)$$

Significance: rescales the three-level HWHM into the *four*-level Γ that appears throughout the LO-dressed Lorentzian $\Lambda(\Omega_{\text{LO}}, \Gamma)$ (Eq. 28, 53, 54). This is the Γ used for Fig. 6/7’s $\Omega_{\text{LO, opt}}$ (Eq. 79) — numerically, with the paper’s stated $\Omega_p/2\pi = 8$ MHz, $\Omega_c/2\pi = 1$ MHz, $\gamma_2/2\pi = 5.2$ MHz, this evaluates to $\Gamma = 7.3255$ MHz, giving $\Omega_{\text{LO, opt}} = 4.2294$ MHz — matching the paper’s stated 4.23 MHz almost exactly. **This Γ is a structurally different quantity from Γ_{FWHM}** used in the LO-free Eq. 17/48/50 — do not conflate the two across figures.

Eq. (79) — the optimal LO bias

$$\Omega_{\text{LO, opt}} = \frac{\sqrt{3}}{3}\Gamma, \quad (\kappa_\rho)_{\text{max}} = \frac{3\sqrt{3}}{8\Gamma}. \quad (79)$$

Significance: found by solving $\partial^2 \Lambda(\Omega_{\text{LO}}, \Gamma)/\partial \Omega_{\text{LO}}^2 = 0$ — the single, physically meaningful bias point that simultaneously maximizes the intrinsic gain coefficient κ_ρ and (by definition of being a stationary point of the second derivative) eliminates second-order distortion. This is the vertical line plotted in Fig. 6(a) and the basis for Fig. 7’s “fixed κ ” curve.

Eq. (80) — practical $P_{\text{out}}(t)$, full derivation chain

$$\begin{aligned} P_{\text{out}}(t) &= P_{\text{in}} e^{-k_p L C_0 \Im(\rho_{21})} \\ &= P_{\text{in}} e^{-k_p L C_0 \bar{A} [1 - \Lambda(\Omega_{\text{LO}}, \Gamma) - \kappa_\rho \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)]} \\ &= P_{\text{in}} e^{-\alpha} e^{\alpha \Lambda(\Omega_{\text{LO}}, \Gamma)} e^{\alpha \kappa_\rho \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)} \\ &= P_{\text{in}} e^{-\alpha} e^{\alpha \Lambda(\Omega_{\text{LO}}, \Gamma)} e^{\kappa \Omega_{\text{RF}} \cos(2\pi \Delta f t + \Delta \phi)}. \end{aligned} \quad (80)$$

Significance: shows step-by-step how substituting Eq. 75 into the Beer–Lambert law (Eq. 2, with $\chi = C_0\rho_{21}$) and factoring out constants leads to the exponential form used just before the final linearization (Eq. 81). Each line is an exact algebraic rearrangement, no approximation yet.

Eq. (81) — linearized practical output

$$P_{\text{out}}(t) \approx P_{\text{in}} e^{-\alpha} e^{\alpha\Lambda(\Omega_{\text{LO}}, \Gamma)} (1 + \alpha\kappa_\rho \Omega_{\text{RF}} \cos(2\pi\Delta f t + \Delta\phi)) = \bar{P}_0 + \kappa \Omega_{\text{RF}} \cos(2\pi\Delta f t + \Delta\phi). \quad (81)$$

Significance: the final approximation step — since $\Omega_{\text{RF}} \ll \Omega_{\text{LO}}$ is assumed, $\alpha\kappa_\rho \Omega_{\text{RF}} \cos(\cdot) \ll 1$, so $e^{(\text{small})} \approx 1 + (\text{small})$. This reproduces Eq. 28 exactly, completing the derivation.

Eq. (82) — refined optimality condition

$$\frac{\partial \kappa}{\partial \Omega_{\text{LO}}} = \frac{\partial(\bar{P}_0 \kappa_\rho)}{\partial \Omega_{\text{LO}}} = \frac{\partial^2 \bar{P}_0}{\partial \Omega_{\text{LO}}^2} = 0. \quad (82)$$

Significance: a more refined optimality condition than Eq. 79’s, which only maximized κ_ρ alone — this one maximizes the full intrinsic gain $\kappa = \alpha\bar{P}_0\kappa_\rho$ directly (i.e., accounts for how $\bar{P}_0$ itself changes with Ω_{LO} , not just κ_ρ).

Eq. (83) — refined optimal LO bias

$$\Omega_{\text{LO}} = \sqrt{\bar{A} - 1 + \sqrt{\bar{A}^2 - 2\bar{A} + 4}} \cdot \frac{\sqrt{3}}{3} \Gamma. \quad (83)$$

Significance: solving Eq. 82 gives this $\bar{A}$ -dependent correction factor multiplying the simple Eq. 79 result. **Not yet numerically checked in this project** against the simple $\Omega_{\text{LO, opt}} = 4.2294$ MHz from Eq. 79 — worth computing if you want to see whether it changes the match to the paper’s stated 4.23 MHz at all (likely a small correction, since $\bar{A}$ is typically not far from a value that makes the correction factor close to 1, but this should be verified numerically, not assumed).

12 Appendix D: Fisher information / SNR derivation

Eq. (84) — two-Lorentzian transmission model

$$\mathcal{T}(\Delta_c; \Omega_{\text{RF}}) = 1 - \mathcal{C} \sum_{i=\pm 1} \frac{(\Gamma_{\text{FWHM}}/2)^2}{(\Delta_c - \frac{\Omega_{\text{RF}}}{2}i)^2 + (\Gamma_{\text{FWHM}}/2)^2}, \quad \mathcal{C} \in (0, 1). \quad (84)$$

Terms: $\mathcal{C}$ = a contrast parameter controlling how deep the two Lorentzian dips are. $i = \pm 1$ sums over the two AT-split peak locations $\Delta_c = \pm\Omega_{\text{RF}}/2$. **Significance: this is the exact model used for Fig. 5** (“use QuTip toolkit to generate the EIT-AT spectrum as two Lorentzian peaks”). **Important detail to verify if revisiting Fig. 5:** this is written as a *transmission-dip* model — 1 minus a sum of two Lorentzian *peaks* in the dip/absorption sense — not a plain additive sum of two Lorentzian *peaks* rising from a zero baseline. If your own Fig. 5 implementation adds two Lorentzians directly (peak-additive form) rather than subtracting them from a constant background, double-check the two forms are actually equivalent for your chosen normalization, since they are not automatically the same thing outside of special cases.

Eq. (85) — Fisher information integral

$$\mathcal{I}(\Omega_{\text{RF}}) = \frac{N_{\text{ph}}}{2\Gamma_{\text{FWHM}}} \int_{-\infty}^{+\infty} \frac{[\partial_{\mathcal{R}} \mathcal{T}(2\Delta_c/\Gamma_{\text{FWHM}}; \mathcal{R})]^2}{\mathcal{T}(2\Delta_c/\Gamma_{\text{FWHM}}; \mathcal{R})} d\left(\frac{2\Delta_c}{\Gamma_{\text{FWHM}}}\right). \quad (85)$$

Terms: N_{ph} = expected photon count within the detection bandwidth (from a Poisson-distributed photon count model at each detuning point). **Significance:** the classical Fisher information for estimating Ω_{RF} (via $\mathcal{R}$) from a full scan of the two-peak transmission spectrum, treating each detuning-point measurement as Poisson-distributed — a rigorous information-theoretic starting point, not a heuristic.

Eq. (86) — Fisher information bounds

$$\mathcal{I}_0(\mathcal{R}) \leq \mathcal{I}(\Omega_{\text{RF}}) \leq \frac{1}{1-2\mathcal{C}} \mathcal{I}_0(\mathcal{R}). \quad (86)$$

Significance: bounds the (generally intractable) exact integral Eq. 85 between a computable closed form $\mathcal{I}_0(\mathcal{R})$ (Eq. 87) and a constant multiple of it — lets the analysis proceed without solving Eq. 85 exactly.

Eq. (87) — closed-form Fisher information lower bound

$$\mathcal{I}_0(\mathcal{R}) = \frac{8N_{\text{ph}}\mathcal{C}^2}{\Gamma_{\text{FWHM}}^2} \frac{1}{1 + \frac{1}{2\mathcal{R}^2}}. \quad (87)$$

Significance: the explicit closed form of the lower bound introduced in Eq. 86 — already has exactly the same $1/(1 + \frac{1}{2\mathcal{R}^2})$ structure that will become $G(\mathcal{R})$ (Eq. 49/90).

Eq. (88) — exact Fisher information with contrast factor

$$\mathcal{I}(\Omega_{\text{RF}}) = \frac{8N_{\text{ph}}\mathcal{C}^2}{\Gamma_{\text{FWHM}}^2} \frac{1}{1 + \frac{1}{2\mathcal{R}^2}} \beta(\mathcal{C}), \quad \beta(\mathcal{C}) \in \left[1, \frac{1}{1-2\mathcal{C}}\right]. \quad (88)$$

Terms: $\beta(\mathcal{C})$ = a contrast-only correction factor, independent of $\mathcal{R}$. **Significance:** the exact Fisher information, shown to differ from the lower-bound Eq. 87 only by the $\mathcal{R}$ -independent factor $\beta(\mathcal{C})$ — for the paper's typical weak-probe simulations ($\mathcal{C} \lesssim 0.2$), $\beta(\mathcal{C}) \leq 1.67$ (≤ 2.22 dB), a small, absorbable correction.

Eq. (89) — Cramér–Rao bound

$$\sigma_{\Omega_{\text{RF}}}^2 \geq \mathcal{I}^{-1}(\Omega_{\text{RF}}) = \frac{\Gamma_{\text{FWHM}}^2}{8N_{\text{ph}}\mathcal{C}^2} \left(1 + \frac{1}{2\mathcal{R}^2}\right). \quad (89)$$

Significance: the standard Cramér–Rao inequality applied to Eq. 88 — gives a lower bound on the variance of any unbiased estimator of Ω_{RF} from the two-Lorentzian spectrum.

Eq. (90) — $G(\mathcal{R})$, derived (not assumed)

$$\frac{\Omega_{\text{RF}}^2}{\sigma_{\Omega_{\text{RF}}}^2} = \frac{\mathcal{R}^2}{1 + \frac{1}{2\mathcal{R}^2}} \equiv G(\mathcal{R}). \quad (90)$$

Significance: multiplying the classical link SNR term $P_{\text{Rx}}|h|^2/\sigma_{\text{Ry}}^2$ by the RF-to-atomic conversion gain $(\wp_{\text{RF}}/\hbar\Gamma_{\text{FWHM}})^2$ and this Fisher-derived factor $G(\mathcal{R})$ yields exactly Eq. 50 — the whole point

of Appendix D is to prove $G(\mathcal{R})$ is a rigorous consequence of information theory applied to AT-splitting estimation, not a prescribed/arbitrary term.

13 Full parameter table (Sec. V-A, as stated in the paper)

Symbol	Value	Meaning
L	1 cm	vapor cell length
N_0	$4.89 \times 10^{10} \text{ cm}^{-3} = 4.89 \times 10^{16} \text{ m}^{-3}$	total atomic density
States	$6S_{1/2} \rightarrow 6P_{3/2} \rightarrow 47D_{5/2} \rightarrow 48P_{3/2}$ (Cs)	4-level ladder
γ_1	0	ground-state decay (none)
$\gamma_2/2\pi$	5.2 MHz	$ 2\rangle$ decay rate
$\gamma_3/2\pi$	3.9 kHz	$ 3\rangle = 47D_{5/2}$ decay rate
$\gamma_4/2\pi$	1.7 kHz	$ 4\rangle = 48P_{3/2}$ decay rate
$\wp_{12}$	$(2.5 ea_0)^2$	$ 1\rangle$ – $ 2\rangle$ dipole moment (already squared)
$\wp_{\text{RF}}$	$-1443.459 ea_0$	$ 3\rangle$ – $ 4\rangle$ dipole moment
f_{RF}	6.9 GHz	RF carrier frequency
λ_p	852 nm	probe wavelength
P_{in}	20.7 μW	stated directly (inconsistent with Eq. 3 given)
d	0.76 mm	probe $1/e^2$ beam diameter
$\Omega_p/2\pi$	8 MHz	probe Rabi frequency
λ_c	510 nm	coupling wavelength
Coupling intensity	17 mW	
d_c	1.95 mm	coupling $1/e^2$ beam diameter
$\Omega_c/2\pi$	1 MHz	coupling Rabi frequency
ϵ_0	$8.854 \times 10^{-12} \text{ F/m}$	vacuum permittivity
Z_0	377 Ω	free-space impedance
$\Omega_{\text{RF}}/2\pi$ (LO-free)	6 MHz	example RF field, LO-free figures
$\Omega_{\text{RF}}/2\pi$ (LO-dressed)	20 kHz	example RF field, LO-dressed figures
$\Omega_{\text{LO}}/2\pi$	4.23 MHz	stated LO-dressed optimum
B	100 kHz	system bandwidth
Δf	15 kHz	LO/RF frequency difference
G_{LNA}	20 dB	LNA gain
R_L	50 Ω	load resistance
D	0.55 A/W	PD responsivity
T	290 K	temperature
η_{eff}	0.5	PD quantum efficiency
G_{Tx}	2.15 dBi	Tx dipole antenna gain
P_{Tx}	30 dBm (1 W)	Tx power
G_{Rx} (conventional only)	2.15 dBi	Rx dipole antenna gain

14 Figure-by-figure equation chains

Fig. 1, Fig. 2 — schematic only

No computation involved.

Fig. 3 — $P_{\text{out}}(\Delta_c, \Omega_{\text{RF}})$, LO-free

Eq. 6 ($\Omega = \Omega_{\text{RF}}$, sweep $\Delta_c, \Omega_{\text{RF}}, \Delta_p = \Delta_{\text{RF}} = 0$) $\rightarrow$ Eq. 7 $\rightarrow$ QuTiP steady state (NOT Eq. 8) $\rightarrow$ Eq. 4 $\rightarrow$ Eq. 2. $P_{\text{in}} = 20.7 \mu\text{W}$ used directly. No Doppler averaging (Eq. 9–11 deliberately excluded).

Fig. 4 — $P_{\text{out}}(\Delta_c, \mathcal{R})$, LO-free

Reuses Fig. 3's grid $\rightarrow$ Eq. 48 to relabel the axis as $\mathcal{R}$. Γ_{FWHM} choice unresolved analytically (see Eq. 77 caution above); this project uses a measured value.

Fig. 5 — $P_{\text{out}}(\Delta_c, d_{\text{Tx-Rx}}) + \text{SNR}$, LO-free

Eq. 15 $\rightarrow$ Eq. 16/17 (needs N_{atoms} , not given) $\rightarrow$ Eq. 13 $\rightarrow$ Eq. 12 $\rightarrow$ Eq. 84 (two-Lorentzian, check dip-vs-peak form) $\rightarrow$ Eq. 50 for the SNR curve.

Fig. 6 — LDR of LO-dressed receiver

Eq. 6 ($\Omega = \Omega_{\text{total}}$) $\rightarrow$ QuTiP steady state $\rightarrow$ Eq. 2/4 $\rightarrow$ real $P_{\text{out}}(\Omega_{\text{total}})$. Overlay Eq. 78/79. Shade LDR using Eq. 57 or Eq. 58 (BOTH need a chosen ϵ). Panel (b): $\Omega_{\text{total}}(t)$ phasor. Panel (c): interpolate panel (a).

Fig. 7 — received-power map, LO-dressed

Same LDR machinery as Fig. 6, translated to P_{Rx} via Eq. 12/13, plotted against $\Omega_{\text{LO}}/2\pi$ (log axis, verified directly from the real figure).

Fig. 8 — SNR vs distance

Eq. 14/15/16/17 $\rightarrow P_{\text{Rx}} \rightarrow$ Eq. 50 / Eq. 52 (or 61) / conventional SNR (using $\lambda_{\text{RF}}^2/4\pi$ as aperture).

Fig. 9 — mutual information vs P_{Rx}

SNR from the Fig. 8 chain $\rightarrow$ Eq. 59 (LO-free) or Eq. 60 (LO-dressed).

Fig. 10 — SER vs P_{Rx}

SNR from the Fig. 8 chain $\rightarrow$ standard M -QAM (conventional, LO-dressed) or M -PAM (LO-free) SER formulas [59].

15 Standing conventions already established in this project

- Γ_{FWHM} (LO-free, Fig. 4/5) = 1.2231 MHz, measured from real QuTiP data, not an analytic candidate.
- Γ (LO-dressed, Fig. 6/7, Eq. 78) = 7.3255 MHz, analytic, gives $\Omega_{\text{LO,opt}} = 4.2294$ MHz (paper: 4.23 MHz).
- N_{atoms} : never given anywhere in the paper or its three cited references.
- ϵ (Eq. 56–58): never given a numeric value. **Both** Eq. 57 and Eq. 58 need it.

- P_{in}/d ($20.7 \mu\text{W} / 0.76 \text{ mm}$) are mutually inconsistent under Eq. 3 (would need $39.08 \mu\text{W}$) — a real, version-confirmed paper-internal inconsistency, introduced in v3.
- Real N_0 makes every QuTiP-computed curve sharper than the paper’s own plots (confirmed recurring across Fig. 3–6; a dedicated audit ruled out Rydberg blockade, Eq. 2/4 form errors, and N_0/L inconsistency as fixable explanations).